

FREEDOM INTERNATIONAL SCHOOL

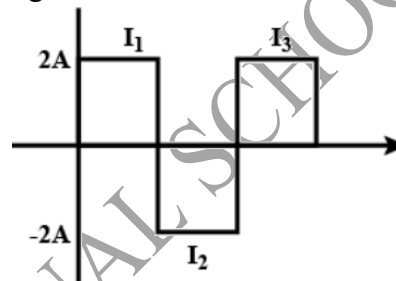
WORKSHEET

PHYSICS

CLASS XII

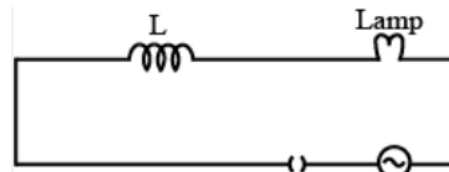
ALTERNATING CURRENT

1. Calculate the rms value of alternating current shown in the figure.



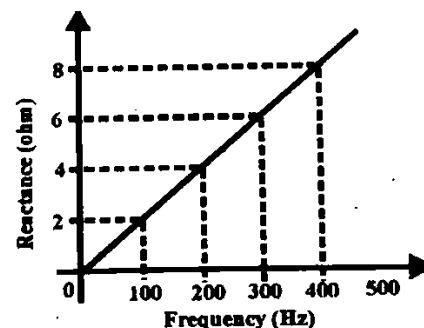
2. The emf of an a.c source is given by the expression $\varepsilon = 300 \sin 314 t$. Write the value of peak voltage and frequency of the source.
3. An ideal inductor of $\frac{5}{\pi}$ H inductance is connected to a 200 V, 50 Hz ac supply.
- (a) Calculate the rms and peak value of current in the inductor.
- (b) What is the phase difference between current through the inductor and the applied voltage? How will it change if a small resistance is connected in series with this inductor in the circuit?

4. An inductor L of reactance X_L , is connected in series with a bulb B to an a.c. source as shown in the figure. Briefly explain how the brightness of the bulb changes, when (i) number of turns of the inductor is reduced and (ii) a capacitor of reactance $X_C = X_L$ is included in series in the same circuit.



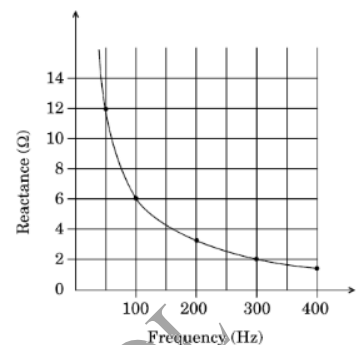
5. Figure shows how the reactance of an inductor varies with frequency.

- (i) Calculate the value of the inductance of the inductor using the information given in the graph.
- (ii) If this inductor is connected in series to a resistor of 8Ω , find what would the impedance be at 300 Hz?



6. A 50 W- 100 V electric bulb is to be used on a 200 V- 50 Hz a.c. supply. Calculate the inductance of the lamp so that it may glow with its normal brightness. (Take $\pi = 3$)
7. The Figure given shows how the reactance of a capacitor varies with frequency

- (i) Use the information from the graph to calculate the value of capacitance of the capacitor.
- (ii) An inductor of inductance 'L' has the same reactance as the capacitor at 100 Hz. Find the value of L.
- (iii) Draw a graph of reactance against frequency for the inductor given in part (ii).
- (iv) If this capacitor and inductor were connected in series to a resistor of $10\ \Omega$, what would the impedance of the combination be at 300 Hz?



8. A town situated 20 km away from a power plant generating power at 440 V, requires 600 kW of electric power at 200 V. The resistance of the two wire line carrying power is $0.4\ \Omega$ per km. The town gets power from the line through a 3000 V- 220 V step down transformer at a substation in the town.
 - (i) Find the line power losses in the form of heat.
 - (ii) How much power must the plant supply, assuming there is negligible power loss due to leakage?
9. Given the input current is 15 A and the input voltage of 100 V for a step- up transformer having 90 % efficiency, find the output power and the voltage in the secondary if the output current is 3 A.
10. A 200 V variable frequency a.c. source is connected to a series combination of $L = 5\text{ H}$, $C = 80\ \mu\text{F}$ and $R = 40\ \Omega$. Calculate (i) angular frequency of the source to get maximum current in the circuit, (ii) the current amplitude at resonance and (iii) the power dissipated in the circuit.